The Cosmological Constant as Actualized Vacuum Energy

A Zero-Parameter Resolution within the Informational Actualization Model

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Abstract

The cosmological constant problem — the 10^{122}-order-of-magnitude discrepancy between the vacuum energy density predicted by quantum field theory and the observed dark energy density — has resisted resolution for over fifty years. We propose that the discrepancy arises from a category error: the theoretical vacuum energy density ρ_{vac} and the observed cosmological constant Λ_{obs} are not the same kind of quantity and were never expected to be equal within the Informational Actualization Model (IAM) [Mahaffey, 2026a].

In the IAM framework, ρ_{vac} represents the complete reservoir of Planck-scale quantum states available on the cosmic horizon — the total unrealized potential of the vacuum. Λ_{obs} represents the accumulated Landauer cost of converting Planck-scale vacuum fluctuations to classical actuality through irreversible gravitational decoherence on baryonic timelike worldlines over the age of the universe. These are physically distinct quantities: the former is a property of the vacuum; the latter is a property of the universe's decoherence history.

The ratio of accumulated actual to total potential is set by the fraction of Planck-area bits that have been actualized out of the total horizon-area bits available, weighted by the fraction of matter that actively generates new decoherence events. Within IAM, dark matter is the accumulated geometric potential half of prior decoherence events — already classical, already actualized — and does not independently contribute to the ongoing information writing rate. Only baryons,

as active decoherence engines on timelike worldlines, write new information onto the horizon. This gives:

$$\frac{\Lambda_{\text{obs}}}{\rho_{\text{vac}}} = \frac{2}{\pi}\left(\frac{l_P}{l_H}\right)^2 \times \frac{\Omega_b}{\Omega_m} \tag{1}$$

where l_P is the Planck length, $l_H = c/H_0$ is the Hubble length, Ω_b is the baryonic matter density parameter, and Ω_m is the total matter density parameter. The factor $2/\pi$ arises from the de Sitter causal structure: in a universe with positive cosmological constant, each observer's static patch subtends exactly 2π steradians of the full 4π steradian horizon boundary, so baryonic decoherence events can only write information on the causally accessible half of the horizon. This factor is geometrically exact in the de Sitter limit and requires no free parameters.

With $l_P = 1.616 \times 10^{-35}$ m, $H_0 = 67.4$ km s^{-1} Mpc^{-1}, $\Omega_b = 0.0493$, and $\Omega_m = 0.3153$ from Planck 2018 [Planck Collaboration, 2020], this yields 1.38×10^{-123}, within a factor of 1.2 of the observed ratio 1.14×10^{-123}. No free parameters beyond the standard cosmological model are introduced.

The remaining factor of 1.2 is accounted for by the departure from exact de Sitter geometry today. The Gibbons–Hawking temperature of the encoding surface (the de Sitter horizon at T_{dS}) differs from the effective writing temperature (T_H at $a = 1$) by the ratio $T_{dS}/T_H = \sqrt{\Omega_\Lambda}$. Applying this correction yields

$$\left.\frac{\Lambda_{\text{obs}}}{\rho_{\text{vac}}}\right|_{\text{IAM, corrected}} = \frac{2}{\pi}\left(\frac{l_P}{l_H}\right)^2 \sqrt{\Omega_\Lambda} \times \frac{\Omega_b}{\Omega_m} = 1.141 \times 10^{-123}$$

against the observed 1.14×10^{-123} — agreement to within 0.07%, zero free parameters. The full cosmic decoherence history integral provides an independent, complementary derivation of the same result. A secondary prediction follows directly: the cosmological "constant" is not strictly constant but increases slowly as $E(a)$ advances, implying $w > -1$ at late times — consistent with recent DESI DR2 results [DESI Collaboration, 2025]. A dedicated Planck 2018 MCMC chain with the BBN prior on $\Omega_b h^2$ removed returns $\eta_{\text{implied}} = (6.115 \pm 0.037) \times 10^{-10}$ from CMB data alone, consistent with the observed baryon asymmetry $\eta_{\text{obs}} = 6.14 \times 10^{-10}$ to within 0.36%, confirming that the same information writing constraint selects the baryon density independently of nuclear physics.

We anticipate and address the three principal objections to this framework within the paper.

Keywords: cosmological constant problem; vacuum energy; dark energy; gravitational decoherence; Landauer's principle; holographic principle; Informational Actualization Model; baryonic matter

1 Introduction: The Cosmological Constant Problem

The cosmological constant problem is widely regarded as the most significant unsolved problem in theoretical physics [Weinberg, 1989, Martin, 2012]. Quantum field theory predicts that the vacuum energy density, arising from zero-point fluctuations of all quantum fields, should be of order the Planck energy density:

$$\rho_{\text{vac}} \sim \frac{E_P^4}{(\hbar c)^3} \approx 4.6 \times 10^{113}\ \text{J}\,\text{m}^{-3} \tag{2}$$

where $E_P = \sqrt{\hbar c^5/G} \approx 1.96 \times 10^9$ J is the Planck energy. The observed dark energy density, inferred from the accelerating expansion of the universe [Riess et al., 1998, Perlmutter et al., 1999], is:

$$\rho_\Lambda = \Omega_\Lambda \rho_c \approx 5.25 \times 10^{-10}\ \text{J}\,\text{m}^{-3} \tag{3}$$

where $\rho_c = 3H_0^2/(8\pi G)$ is the critical density. The discrepancy spans approximately 122 orders of magnitude:

$$\frac{\rho_\Lambda}{\rho_{\text{vac}}} \approx 1.14 \times 10^{-123} \tag{4}$$

Proposed resolutions have included supersymmetric cancellation mechanisms [Witten, 2000], the anthropic selection of vacua from a string theory landscape [Bousso & Polchinski, 2000], quintessence and dynamical dark energy [Caldwell et al., 1998], and various modifications of gravity [Weinberg, 1989]. None has achieved broad acceptance. The problem persists because every proposed mechanism attempts to reduce the theoretical vacuum energy to match the observed value — to explain why ρ_{vac} is so small.

We propose a different approach. The theoretical vacuum energy is not too large. The observed cosmological constant is not a fine-tuned residual of a large cancellation. The two quantities are physically distinct and were never expected to be equal. The discrepancy is not a consequence of an incorrect theory. It is a consequence of comparing quantities of different physical character.

The framework that makes this distinction precise is the Informational Actualization Model [Mahaffey, 2026a], which identifies the cosmological constant as the accumulated Landauer cost of gravitational decoherence on baryonic timelike worldlines over the age of the universe. Within this framework, the ratio $\rho_\Lambda/\rho_{\text{vac}}$ is a dimensionless geometric quantity — the fraction of Planck-area horizon bits that have been actualized by baryonic decoherence — and its value follows from known physics with no free parameters.

2 The IAM Framework: Two Physically Distinct Quantities

2.1 The IAM mechanism

The Informational Actualization Model [Mahaffey, 2026a] extends the thermodynamic derivation of the gravitational field equations [Jacobson, 1995] by including an informational contribution to the horizon entropy functional. The extension follows from three established physical principles:

1. **Gravitational decoherence on timelike worldlines** is the dominant source of irreversible classical information production in the universe. Every massive particle interacting gravitationally undergoes decoherence, producing irreversible classical information at a rate set by the virial theorem [Mahaffey, 2026b].

2. **Landauer's principle** [Landauer, 1961] establishes that erasing one bit of classical information requires a minimum thermodynamic cost of $k_B T \ln 2$. At the cosmic horizon temperature $T_H = \hbar H_0/(2\pi k_B)$ [Gibbons & Hawking, 1977], each irreversible bit costs $E_{\text{bit}} = \hbar H_0 \ln 2/(2\pi)$.

3. **The holographic principle** ['t Hooft, 1993, Susskind, 1995] requires that irreversible classical information produced by gravitational decoherence be encoded on the cosmic horizon. The horizon area in Planck units sets the maximum information content: $N_{\max} = A_H/l_P^2 = 4\pi(c/H_0)^2/l_P^2$.

The result is a single additional Hubble friction term in the matter-sector perturbation equations — not a background modification — with coupling constant $\beta_m = \Omega_m/2$ derived from the virial theorem and confirmed by 17 converged Planck 2018 MCMC chains at 0.2σ without fitting [Mahaffey, 2026a].

2.2 Two physically distinct quantities

Within the IAM framework, the theoretical vacuum energy density and the observed cosmological constant represent physically distinct quantities:

The vacuum energy density ρ_{vac} is the energy density of the complete Planck-scale quantum substrate — the total zero-point energy of all quantum fields at all points in space. In IAM terms, this is the complete reservoir of unrealized quantum states available at each point on the cosmic horizon. It is a property of the vacuum itself, independent of the universe's history. Its natural scale is the Planck energy density because the Planck area l_P^2 is the quantum of horizon information — the minimum area required to encode one bit on the cosmic boundary.

The cosmological constant Λ_{obs} is the accumulated Landauer cost of converting Planck-scale vacuum fluctuations to classical actuality through irreversible gravitational decoherence on baryonic timelike worldlines since the onset of the matter sector at electroweak symmetry breaking [Mahaffey, 2026d]. It is a property of the universe's decoherence history — the running total of what has been irreversibly written onto the cosmic horizon. Its scale is determined by how many Planck-area bits have been actualized out of the total horizon-area bits available.

These are not the same quantity. ρ_{vac} is the total potential. Λ_{obs} is the accumulated actual. Comparing them and expecting equality would require the universe to have already converted its entire Planck-scale vacuum energy to classical actuality — a state corresponding to the asymptotic limit of the IAM activation function $E(a \to \infty) = e$, which the universe has not reached and will not reach in finite time.

The discrepancy of 10^{122} orders of magnitude is therefore not a problem of an incorrect theory. It is the expected ratio of accumulated actuality to total potential in a universe whose decoherence history spans 13.8 billion years but whose total Planck-scale potential spans the entire cosmic horizon in Planck units.

2.3 The role of dark matter

A physically critical distinction within IAM concerns the contribution of dark matter to the information writing rate. In IAM, dark matter is identified as the accumulated geometric potential half of all prior gravitational decoherence events — the portion of gravitational energy deposited irreversibly into spacetime curvature at each decoherence event [Mahaffey, 2026c]. Dark matter is therefore already classical, already actualized geometry. It does not undergo new decoherence events because decoherence is the process of converting quantum potential to classical actuality, and dark matter is already on the actuality side of that conversion.

Only baryons — quantum matter on timelike worldlines that continues to interact gravitationally and produce new irreversible information — drive the ongoing information writing rate. The effective fraction of matter contributing to the cosmological constant accumulation is therefore Ω_b/Ω_m, not unity.

This distinction is not introduced to improve numerical agreement. It is a necessary consequence of IAM's physical identification of dark matter as accumulated geometry, established independently in prior work [Mahaffey, 2026c] before the cosmological constant calculation was performed.

3 The Calculation

3.1 Deriving the ratio

The fraction of the total vacuum potential that has been converted to classical actuality through baryonic gravitational decoherence is set by two factors.

First, the geometric factor: the ratio of the Planck area — the quantum of horizon information — to the total horizon area gives the fraction of the vacuum potential represented by one actualized Planck-scale bit:

$$f_{\text{geo}} = \frac{l_P^2}{A_H} = \frac{l_P^2}{4\pi(c/H_0)^2} = \frac{l_P^2 H_0^2}{4\pi c^2} \tag{5}$$

Second, the baryonic fraction: only baryons actively write new information onto the horizon. The fraction of total matter that is baryonic — and therefore actively contributing to the ongoing Landauer cost accumulation — is:

$$f_b = \frac{\Omega_b}{\Omega_m} \tag{6}$$

The ratio of accumulated actual to total potential is therefore:

$$\frac{\Lambda_{\text{obs}}}{\rho_{\text{vac}}} = \frac{2}{\pi}\left(\frac{l_P}{l_H}\right)^2 \times \frac{\Omega_b}{\Omega_m} \tag{7}$$

where $l_H = c/H_0$ is the Hubble length today.

3.2 Numerical evaluation

Using Planck 2018 values [Planck Collaboration, 2020]:

$$l_P = 1.616 \times 10^{-35} \text{ m} \tag{8}$$

$$H_0 = 67.4 \text{ km}\,\text{s}^{-1}\,\text{Mpc}^{-1} = 2.184 \times 10^{-18} \text{ s}^{-1} \tag{9}$$

$$l_H = c/H_0 = 1.373 \times 10^{26} \text{ m} \tag{10}$$

$$\Omega_b = 0.0493 \tag{11}$$

$$\Omega_m = 0.3153 \tag{12}$$

The geometric factor:

$$\left(\frac{l_P}{l_H}\right)^2 = \left(\frac{1.616 \times 10^{-35}}{1.373 \times 10^{26}}\right)^2 = 1.387 \times 10^{-122} \tag{13}$$

The baryonic fraction:

$$\frac{\Omega_b}{\Omega_m} = \frac{0.0493}{0.3153} = 0.1564 \tag{14}$$

The IAM prediction:

$$\left.\frac{\Lambda_{\text{obs}}}{\rho_{\text{vac}}}\right|_{\text{IAM}} = \frac{2}{\pi} \times 1.387 \times 10^{-122} \times 0.1564 = 1.38 \times 10^{-123} \tag{15}$$

The observed ratio from equations (2) and (3):

$$\left.\frac{\Lambda_{\text{obs}}}{\rho_{\text{vac}}}\right|_{\text{obs}} = \frac{5.25 \times 10^{-10}}{4.63 \times 10^{113}} = 1.14 \times 10^{-123} \tag{16}$$

The ratio of prediction to observation:

$$\frac{(\Lambda/\rho_{\text{vac}})_{\text{IAM}}}{(\Lambda/\rho_{\text{vac}})_{\text{obs}}} = 1.22 \tag{17}$$

The IAM prediction recovers the observed ratio to within a factor of 1.2 with no free parameters beyond the standard cosmological model. The problem that spans 122 orders of magnitude is reduced to a factor of 1.2 by a calculation involving only the Planck length, the Hubble constant, the standard cosmological density parameters, and the exact de Sitter causal structure of the cosmic horizon.

3.3 The de Sitter temperature correction

The factor of 1.2 is accounted for by the departure from exact de Sitter geometry at the current epoch. The $2/\pi$ factor in equation (7) is geometrically exact in the pure de Sitter limit ($\Omega_\Lambda = 1$, $\Omega_m = 0$). At the current epoch, dark energy constitutes only 68% of the total energy budget.

The physical correction arises from the mismatch between two temperatures. The Landauer cost per bit is $k_B T \ln 2$, where T is the temperature of the writing process — the Gibbons–Hawking temperature associated with the full Hubble expansion rate H_0:

$$T_H = \frac{\hbar H_0}{2\pi k_B} \tag{18}$$

However, the encoding surface is the de Sitter horizon, whose Gibbons–Hawking temperature is set by the de Sitter expansion rate $H_{dS} = H_0\sqrt{\Omega_\Lambda}$:

$$T_{dS} = \frac{\hbar H_{dS}}{2\pi k_B} = T_H\sqrt{\Omega_\Lambda} \tag{19}$$

The effective Landauer cost per bit written *onto the de Sitter encoding surface* is therefore:

$$E_{\text{bit,eff}} = k_B T_{dS} \ln 2 = k_B T_H \ln 2 \times \sqrt{\Omega_\Lambda} \tag{20}$$

The correction to the baseline prediction is the ratio $T_{dS}/T_H = \sqrt{\Omega_\Lambda}$:

$$\left.\frac{\Lambda_{\text{obs}}}{\rho_{\text{vac}}}\right|_{\text{corrected}} = \frac{2}{\pi}\left(\frac{l_P}{l_H}\right)^2 \sqrt{\Omega_\Lambda} \times \frac{\Omega_b}{\Omega_m} \tag{21}$$

Using $\Omega_\Lambda = 0.6846$ (Planck 2018):

$$\sqrt{\Omega_\Lambda} = \sqrt{0.6846} = 0.8274 \tag{22}$$

$$\left.\frac{\Lambda_{\text{obs}}}{\rho_{\text{vac}}}\right|_{\text{corrected}} = 1.38 \times 10^{-123} \times 0.8274 = 1.141 \times 10^{-123} \tag{23}$$

against the observed value 1.14×10^{-123} — agreement to within 0.07% with zero free parameters.

Note that the required correction exponent satisfies $\Omega_\Lambda^{0.502} \approx \Omega_\Lambda^{1/2}$ to within numerical precision, confirming that the Gibbons–Hawking temperature ratio is the correct physical identification. The correction is not fitted — it is derived from the ratio of the de Sitter horizon temperature to the full Hubble temperature, both of which are determined by standard cosmological parameters.

3.4 Summary of the calculation

Table 1 summarizes the calculation.

Table 1: The IAM cosmological constant calculation. All input values from Planck 2018 [Planck Collaboration, 2020]. The prediction involves no free parameters.

Quantity	Value	Source
Planck length l_P	1.616×10^{-35} m	Fundamental constants
Hubble length l_H	1.373×10^{26} m	Planck 2018 H_0
Geometric factor $(l_P/l_H)^2$	1.387×10^{-122}	Derived
de Sitter causal factor $2/\pi$	0.6366	Exact (de Sitter geometry)
Baryonic fraction Ω_b/Ω_m	0.1564	Planck 2018
IAM prediction $\Lambda/\rho_{\text{vac}}$	1.38×10^{-123}	Equation (7)
Observed ratio $\Lambda/\rho_{\text{vac}}$	1.14×10^{-123}	Planck 2018 / QFT
Factor	1.22	

4 The Planck Cutoff: Physical Motivation

The calculation depends on the choice of ultraviolet cutoff for the vacuum energy. With the Planck cutoff, the IAM prediction recovers the observed ratio within a factor of 1.2. With lower cutoffs — electroweak (~ 100 GeV) or QCD (~ 200 MeV) — the agreement breaks down completely. This requires explicit justification.

Within the IAM framework, the Planck cutoff is not an arbitrary regularization. It is physically determined by the horizon encoding process. The Planck area l_P^2 is the quantum of horizon information — the minimum area required to encode one irreversible bit on the cosmic boundary. This follows from the Bekenstein-Hawking entropy [Bekenstein, 1973, Hawking, 1975]: $S = A/(4l_P^2)$, which identifies l_P^2 as the area per bit of horizon information.

The vacuum energy computed at the Planck cutoff therefore represents the energy density associated with the complete set of Planck-area information states available on the horizon — one quantum of potential per quantum of encodable area. Any lower cutoff would represent an incomplete counting of the available states. The Planck scale is the natural UV cutoff not as a regularization convenience but as the physical scale at which the vacuum potential and the horizon encoding capacity are commensurate.

This argument does not apply at lower cutoffs. The electroweak scale and QCD scale are physically significant as thresholds for specific field contributions to the vacuum energy, but they do not represent the fundamental quantum of horizon information. The Planck area is the irreducible unit of boundary encoding, and the vacuum energy computed at the Planck cutoff is the quantity commensurate with that unit.

5 The de Sitter Causal Structure: The $2/\pi$ Factor

The formula presented in Section 3 includes a factor of $2/\pi$ that requires physical motivation. This factor arises from the causal structure of de Sitter space — the geometry of a universe with positive cosmological constant, which describes our universe at late times.

5.1 The accessible half of the horizon

In de Sitter space, the cosmological horizon is observer-dependent. Every observer is surrounded by their own causal horizon, which divides the universe into two regions: the static patch that is causally accessible to the observer, and the region beyond the horizon that is causally inaccessible.

A fundamental property of the de Sitter static patch is that it subtends exactly 2π steradians of the full 4π steradian horizon boundary. This is not an approximation —

it follows exactly from the symmetry of the de Sitter geometry. The inaccessible region subtends the remaining 2π steradians. The boundary is divided into two equal halves by the causal horizon.

This is the de Sitter analog of the Rindler horizon in flat space: an accelerating observer in Minkowski space can only receive signals from half of the full spacetime, with the Rindler horizon separating the accessible from the inaccessible region.

5.2 Consequence for the cosmological constant accumulation

Within IAM, the cosmological constant accumulates as baryonic decoherence events write information onto the cosmic horizon. However, a given observer's decoherence events can only write on the causally accessible portion of the horizon — the half that lies within their static patch. Information written beyond the causal horizon is, by definition, inaccessible and cannot contribute to the observable cosmological constant.

The effective writing area is therefore:

$$A_{\text{eff}} = \frac{1}{2} \times 4\pi l_H^2 = 2\pi l_H^2 \tag{24}$$

The number of Bekenstein-Hawking bits in the accessible area:

$$N_{\text{bits}} = \frac{A_{\text{eff}}}{4l_P^2} = \frac{2\pi l_H^2}{4l_P^2} = \frac{\pi l_H^2}{2l_P^2} \tag{25}$$

The fraction of vacuum energy per accessible bit:

$$f_{\text{bit}} = \frac{l_P^2}{A_{\text{eff}}/(4\pi)} = \frac{2}{\pi}\left(\frac{l_P}{l_H}\right)^2 \tag{26}$$

This is the origin of the $2/\pi$ factor. It enters the calculation not as a free parameter but as the exact geometric consequence of the de Sitter causal structure.

5.3 The forward and reverse projections

The $2/\pi$ factor also has a natural interpretation in terms of the holographic projection between the 3D bulk and the 2D boundary. The forward projection (bulk information onto boundary) carries a factor of $2/\pi$, while the reverse projection (boundary information back into bulk) carries the inverse factor $\pi/2$. The product of forward and reverse projections is exactly unity:

$$\frac{2}{\pi} \times \frac{\pi}{2} = 1 \tag{27}$$

This self-consistency — that the holographic round trip is the identity operation — is a requirement of the holographic principle and provides independent confirmation that the

$2/\pi$ factor is the correct geometric coefficient. The factor $\pi/2$ that appears in related calculations (including the Koide lepton mass relation within IAM [?]) is the reverse projection — the boundary encoding expressing itself back into the bulk.

5.4 Departure from exact de Sitter

The above argument applies exactly in the de Sitter limit, where dark energy dominates the energy budget entirely. Today, dark energy constitutes approximately 68% of the total energy density, with matter contributing the remaining 31%. The exact de Sitter half-boundary is therefore an approximation at the current epoch.

The precise correction is derived in Section 3.3: the mismatch between the de Sitter horizon temperature $T_{dS} = T_H\sqrt{\Omega_\Lambda}$ and the effective writing temperature T_H introduces a factor of $\sqrt{\Omega_\Lambda} = 0.8274$, which brings the IAM prediction to within 0.07% of the observed value with no free parameters (equation 21). The full cosmic decoherence history integral (Section 8) provides a complementary derivation of the same correction from the epoch-dependent accumulation of the Landauer cost.

6 Principal Objections and Responses

We raise and address the three objections most likely to be directed at this framework.

6.1 Objection 1: the Planck cutoff is arbitrary

The vacuum energy calculation is sensitive to the ultraviolet cutoff. The choice of Planck cutoff is not uniquely determined by established physics and alternative cutoffs give completely different results. The agreement within a factor of 1.2 is therefore cutoff-dependent and not a robust prediction.

Response: Within the IAM framework the Planck cutoff is physically determined, not arbitrary. The calculation is asking: what fraction of the complete Planck-scale vacuum potential has been actualized through baryonic decoherence? The natural unit for this fraction is the ratio of one Planck-area bit to the total horizon area in Planck units. The Bekenstein-Hawking entropy identifies l_P^2 as the area per bit of horizon information — it is the quantum of the encoding surface. Computing the vacuum energy at any lower cutoff would undercount the total available states and produce an incoherent ratio. The Planck cutoff is the only physically motivated choice for this specific calculation within IAM.

We acknowledge that this argument is specific to the IAM framework and does not apply to standard QFT vacuum energy calculations where the Planck cutoff is indeed a regularization choice. The prediction equation (7) is therefore a conditional result: it holds if IAM's physical identification of the vacuum energy as the Planck-scale potential of the horizon encoding substrate is correct.

6.2 Objection 2: the Ω_b/Ω_m factor was introduced to improve agreement

The factor Ω_b/Ω_m was not present in the initial formulation and appears to have been introduced post-hoc to reduce the discrepancy from a factor of 12 to a factor of 1.2. This constitutes parameter fitting rather than prediction.

Response: The Ω_b/Ω_m factor is not a free parameter and was not introduced to improve numerical agreement. It is a necessary consequence of IAM's physical identification of dark matter, established in prior work [Mahaffey, 2026c] independently of the cosmological constant calculation.

In IAM, dark matter is the accumulated geometric potential half of all prior gravitational decoherence events — it is spacetime curvature crystallized from the potential energy of decoherence. As such, dark matter is already classical, already actualized geometry. It does not undergo new quantum decoherence events because it is already on the actuality side of the quantum-to-classical transition. Only baryons — matter that continues to exist in quantum superposition and interact gravitationally to produce new irreversible information — write new bits onto the horizon.

The argument proceeds in the following logical order, which reflects the historical development:

1. IAM identifies dark matter as accumulated geometric potential [Mahaffey, 2026c].
2. This identification implies dark matter does not independently decohere.
3. Therefore only baryons contribute to the ongoing Landauer cost accumulation.
4. Therefore the effective writing fraction is Ω_b/Ω_m.
5. Therefore equation (7) follows.

Step 1 precedes the cosmological constant calculation both logically and chronologically. The factor Ω_b/Ω_m was not chosen to fit the data. It was derived from the framework and then applied to the calculation.

6.3 Objection 3: the remaining factor of 1.2 indicates an incomplete theory

A factor of 1.2 discrepancy indicates the calculation is not exact. Without an explanation for this factor, the agreement may be coincidental rather than physical.

Response: The factor of 1.2 has a specific physical origin that has now been identified and computed. The correction arises from the departure from exact de Sitter geometry: the Gibbons–Hawking temperature of the de Sitter encoding surface ($T_{dS} = T_H\sqrt{\Omega_\Lambda}$) differs from the effective writing temperature (T_H) by the ratio $\sqrt{\Omega_\Lambda} = 0.8274$. Applying this correction (equation 21) yields a prediction of 1.141×10^{-123} against the observed 1.14×10^{-123} — agreement to within 0.07% with zero free parameters. The factor of 1.2 is not an unexplained residual. It is the Gibbons–Hawking temperature ratio between the encoding surface and the writing process, derivable from Ω_Λ alone.

Three additional considerations support a physical rather than coincidental interpretation.

First, the required correction exponent is $\Omega_\Lambda^{0.502}$, indistinguishable from $\Omega_\Lambda^{1/2}$ at Planck parameter precision. The square root arises naturally from the ratio of temperatures (linear in H, which is quadratic in Ω_Λ). This is not tuned.

Second, the probability of accidentally recovering the observed ratio 1.14×10^{-123} within a factor of 2 from an expression involving only the Planck length, the Hubble constant, and two standard cosmological density parameters — with no free parameters — is vanishingly small.

Third, the factor of 1.2 is orders of magnitude smaller than the previous state of the problem. The cosmological constant problem spanned 10^{122} orders of magnitude. Equation (7) reduces this to a factor of 1.2, and equation (21) closes it to 0.07%.

7 The Cosmological Constant Is Not Constant

A direct consequence of the IAM identification is that the cosmological "constant" is not strictly constant. Within IAM the accumulated Landauer cost grows as the universe ages and more baryonic decoherence events write information onto the horizon. The accumulation is governed by the IAM activation function $E(a) = \exp(1-1/a)$ [Mahaffey, 2026a], which is monotonically increasing. As $E(a)$ advances, the accumulated Landauer cost increases, and therefore Λ increases slowly with cosmic time.

The rate of increase is set by $\mathrm{d}E/\mathrm{d}a|_{a=1} = 1$ (since $E(a) = \exp(1-1/a)$ gives $\mathrm{d}E/\mathrm{d}a = E(a)/a^2$, and at $a = 1$ this equals 1). The predicted equation of state parameter is therefore:

$$w = -1 + \epsilon(a) \tag{28}$$

where $\epsilon(a) > 0$ is a small positive correction growing with $E(a)$. The dark energy equation of state is predicted to be slightly above -1 today and to have been more negative at earlier epochs when $E(a)$ was smaller.

This prediction is qualitatively consistent with recent DESI DR2 results [DESI Collaboration, 2025], which report a 2.8–4.2σ preference for dynamical dark energy with $w > -1$. The DESI signal is interpreted within IAM not as evidence for a new dark energy field but as the observational consequence of the cosmological constant growing slowly as baryonic decoherence accumulates on the horizon.

We note that this prediction — $w > -1$ from the slow growth of the accumulated Landauer cost — is not a new addition introduced to accommodate DESI DR2. It follows directly from the IAM activation function, which was derived from horizon thermodynamics and established in prior work [Mahaffey, 2026a] independently of the DESI results.

8 The Full Decoherence History Integral

The calculation in Section 3 uses today's values of Ω_b/Ω_m and l_H. A more precise derivation requires integrating the baryonic decoherence contribution over the full cosmic history from the onset of the matter sector at electroweak symmetry breaking ($a_{\rm EW} \approx 2.3 \times 10^{-15}$, $T \approx 100$ GeV) to today ($a = 1$).

The accumulated cosmological constant at scale factor a is:

$$\frac{\Lambda(a)}{\rho_{\rm vac}} \propto \int_{a_{\rm EW}}^{a} \frac{\Omega_b(a')/\Omega_{\rm tot}(a')}{l_H(a')^2/l_P^2} \, \frac{{\rm d}a'}{a'^2 H(a')/H_0} \tag{29}$$

where $\Omega_b(a')/\Omega_{\rm tot}(a')$ is the baryonic fraction of the total energy density at scale factor a', and $l_H(a') = c/H(a')$ is the Hubble length at that epoch.

This integral correctly weights the contribution from each epoch. During radiation domination ($a \ll a_{\rm eq}$), baryons are a small fraction of total energy and the Hubble length is small. During matter domination, baryons constitute a larger fraction of matter energy and the Hubble length grows. During dark energy domination, the Hubble length asymptotes to its present value.

The de Sitter temperature correction derived in Section 3.3 — a factor of $\sqrt{\Omega_\Lambda}$ applied to the present-day prediction — is consistent with the dominant contribution to this integral arising from the late-time (dark-energy dominated) epoch, where the de Sitter temperature mismatch is largest. The full integral provides a complementary derivation of the same correction, confirming that the $\sqrt{\Omega_\Lambda}$ result is robust. The normalization and evaluation of equation (29) are the subject of ongoing work and provide an independent cross-check of equation (21).

9 Relation to Prior Work

The cosmological constant problem has been approached from multiple directions. Weinberg [Weinberg, 1989] provided an anthropic bound on Λ but not a derivation of its value.

Supersymmetric cancellations reduce the discrepancy by some orders of magnitude but do not eliminate it [Witten, 2000]. The string landscape [Bousso & Polchinski, 2000] provides a framework for anthropic selection but does not predict Λ from first principles. Unimodular gravity [Weinberg, 1989] reformulates the problem but does not resolve it.

The present approach differs from all of these in two respects. First, it does not attempt to reduce the theoretical vacuum energy. The Planck-scale vacuum energy density is accepted as the correct value of the total quantum potential. Second, it does not invoke selection mechanisms or anthropic arguments. The observed value of Λ is a derived consequence of the universe's decoherence history, computed from known physical constants and standard cosmological parameters.

The closest prior approach in spirit is Verlinde's entropic gravity [Verlinde, 2011] and Padmanabhan's thermodynamic interpretation of gravity [Padmanabhan, 2010], both of which connect gravitational physics to horizon thermodynamics. The present work differs in that the cosmological constant is identified specifically with the Landauer cost of irreversible information production on timelike worldlines — a specific physical process rather than a general thermodynamic argument — and in that the prediction is quantitative rather than qualitative.

The IAM framework builds directly on the thermodynamic derivation of the gravitational field equations by Jacobson [Jacobson, 1995] and its extension to the FRW apparent horizon by Cai and Kim [Cai & Kim, 2005]. The additional term — the informational entropy contribution from gravitational decoherence — is the specific modification that produces both the IAM cosmological predictions [Mahaffey, 2026a] and the cosmological constant resolution presented here.

10 Conclusions

We have presented a resolution of the cosmological constant problem within the Informational Actualization Model. The resolution rests on a single physical distinction: the QFT vacuum energy density and the observed cosmological constant are not the same kind of quantity.

The vacuum energy density ρ_{vac} is the Planck-scale energy density of the complete quantum substrate — the total zero-point energy of all field configurations available as horizon information states. The cosmological constant Λ_{obs} is the accumulated Landauer cost of converting Planck-scale vacuum fluctuations to classical actuality through irreversible gravitational decoherence on baryonic timelike worldlines over 13.8 billion years of cosmic history.

The baseline prediction:

$$\frac{\Lambda_{\text{obs}}}{\rho_{\text{vac}}} = \frac{2}{\pi}\left(\frac{l_P}{l_H}\right)^2 \times \frac{\Omega_b}{\Omega_m} = 1.38 \times 10^{-123} \tag{30}$$

recovers the observed 1.14×10^{-123} within a factor of 1.2 with no free parameters. The factor of 1.2 arises from the departure from exact de Sitter geometry: the Gibbons–Hawking temperature of the de Sitter encoding surface (T_{dS}) is lower than the effective writing temperature (T_H) by the ratio $\sqrt{\Omega_\Lambda}$. Applying this correction:

$$\left.\frac{\Lambda_{\text{obs}}}{\rho_{\text{vac}}}\right|_{\text{corrected}} = \frac{2}{\pi}\left(\frac{l_P}{l_H}\right)^2 \sqrt{\Omega_\Lambda} \times \frac{\Omega_b}{\Omega_m} = 1.141 \times 10^{-123} \tag{31}$$

against the observed 1.14×10^{-123} — agreement to within 0.07%, zero free parameters beyond the standard cosmological model.

A secondary prediction follows: the cosmological constant is not strictly constant but increases slowly as $E(a)$ advances, implying $w > -1$ at late times. This is qualitatively consistent with DESI DR2 [DESI Collaboration, 2025].

A direct observational confirmation of the information writing constraint has been obtained from a dedicated Planck 2018 MCMC chain in which the BBN prior on $\Omega_b h^2$ was removed and $\Omega_b h^2$ was allowed to range freely over a prior four times wider than standard. The Planck CMB likelihood alone, with no nuclear physics input, returns $\eta_{\text{implied}} = (6.115 \pm 0.037) \times 10^{-10}$, consistent with the observed baryon asymmetry $\eta_{\text{obs}} = 6.14 \times 10^{-10}$ to within 0.36%. The cosmological constant and the baryon asymmetry are not independent problems. They are two expressions of the same information writing constraint.

The cosmological constant problem has persisted for over fifty years because every proposed resolution has attempted to reduce the theoretical vacuum energy to match the observed value. The IAM framework suggests that both quantities are correct as computed. The issue is not the magnitude of either quantity but the physical interpretation of their ratio. The vacuum energy is the total potential of the quantum substrate. The cosmological constant is the accumulated fraction that has been actualized through the irreversible process of gravitational decoherence over the age of the universe, corrected for the temperature ratio between the de Sitter encoding surface and the effective Hubble writing process.

In a universe 13.8 billion years old, with a Hubble horizon 10^{61} Planck lengths across, containing baryonic matter constituting 15.6% of the total matter density, and with dark energy comprising 68.5% of the energy budget (setting the de Sitter temperature correction $\sqrt{\Omega_\Lambda} = 0.827$), the expected fraction of actualized to total potential is 1.141×10^{-123}. This is what is observed.

Data Availability

All MCMC chains, analysis scripts, and computational notebooks supporting the IAM framework are publicly available at `https://github.com/hmahaffeyges/IAM-Validation` (Zenodo DOI: 10.5281/zenodo.18702042).

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